TMUA · JANUARY 2027 COHORT · SET B

Algebra & functions

Companion practice · a step below the challenge set

Fifteen original questions with the same topic sequence as the harder pack, but fewer interacting conditions and cleaner calculations. The emphasis remains on choosing an efficient method and keeping every restriction.

15 ORIGINAL QUESTIONS	NO CALCULATOR	SUGGESTED: 45 MINUTES

Use this set before the challenge questions

Attempt the questions independently, then make a second pass without a time limit. Use the worked answers only after recording your choices. A question's number matches the topic of the same-numbered question in the harder pack.

QUESTIONS	COVERAGE
01–02	Fractional indices, reciprocal structure, exact surds and integer bounds.
03–06	Quadratic root location, positivity, simultaneous equations and rational inequalities.
07–08	Non-monic divisors, factor and remainder theorems, and polynomial division.
09–13	Composition, radicals, modulus, rational ranges and transformed quadratics.
14–15	Invalid algebraic steps and necessary/sufficient conditions.

Instructions. Choose exactly one answer for each question. All square roots denote principal, non-negative roots. "Distinct" means different values, not repeated occurrences of a root. Unless otherwise stated, variables are real.

Scope. This is a questions-only companion to the existing concept notes. Sequences, series and binomial expansion are excluded; ordinary polynomial expansion and factorisation remain in scope. The difficulty and timing are teaching choices, not official ratings.

NAME: _____ DATE: _____

Your answers

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15

Mark any answer you guessed. After review, record the restriction or key observation that changed your approach.

Independent teaching material prepared for Thriving Scholars' January 2027 cohort. Not an official UAT-UK paper and not endorsed by UAT-UK.

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 1–2

QUESTION 01

Choose one answer.

The positive real number x satisfies

$$x^{2/3} + x^{-2/3} = 7.$$

What is the sum of all the distinct possible values of x?

A 7

B 9

C 18

D 21

E 27

F 49

Working

QUESTION 02

Choose one answer.

How many non-negative integers n satisfy

$$\sqrt{n+5} - \sqrt{n+1} > \frac{1}{2}\,?$$

A 11

B 12

C 13

D 14

E 15

F 16

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 3–4

QUESTION 03

Choose one answer.

For which real values of k does the equation

$$x^2 - 6x + k = 0$$

have two distinct real roots, both strictly greater than 1?

A $k < 9$

B $0 < k < 9$

C $5 \leq k < 9$

D $5 < k \leq 9$

E $5 < k < 9$

F $k > 5$

Working

QUESTION 04

Choose one answer.

For which real values of k is

$$(k-1)x^2 + 2(k-1)x + 3 > 0$$

true for every real value of x?

A $1 \leq k < 4$

B $1 < k < 4$

C $1 \leq k \leq 4$

D $k < 4$

E $0 \leq k < 4$

F $k > 1$

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 5–6

QUESTION 05

Choose one answer.

The line $y = mx + c$ meets the parabola $y = x^2 - 2x$ at two distinct points P and Q.

The midpoint of PQ has x-coordinate 3, and the positive difference between the x-coordinates of P and Q is 4.

What is the value of c?

A -9

B -5

C -1

D 1

E 5

F 9

Working

QUESTION 06

Choose one answer.

What is the complete solution set of

$$\frac{(x-2)(x+1)^2}{(x-2)(x-1)} \geq 0\,?$$

A $(1, \infty)$

B $\{-1\} \cup (1, \infty)$

C $\{-1\} \cup [1, 2) \cup (2, \infty)$

D $(-\infty, 1) \setminus \{-1\}$

E $(1, 2) \cup (2, \infty)$

F $\{-1\} \cup (1, 2) \cup (2, \infty)$

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 7–8

QUESTION 07

Choose one answer.

A polynomial $P(x)$ leaves remainder 4 when divided by $2x - 1$, and remainder 1 when divided by $x + 1$.

What is the remainder when $P(x)$ is divided by $2x^2 + x - 1$?

A $2x + 3$

B $3x + 2$

C $x + 3$

D $2x + 1$

E 4

F 3

Working

QUESTION 08

Choose one answer.

What is the remainder when

$$4x^4 + 2x^3 + 3x + 1$$

is divided by $2x^2 - x - 1$?

A $5x + 1$

B $5x + 3$

C $7x + 1$

D $7x + 3$

E $3x + 7$

F $7x - 3$

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 9–10

QUESTION 09

Choose one answer.

The function f is defined for all real x by

$$f(x) = x^2 - 2x.$$

How many distinct real solutions does $f(f(x)) = 3$ have?

A 1

B 2

C 3

D 4

E 5

Working

QUESTION 10

Choose one answer.

For which real values of a does the equation

$$\sqrt{x} = x + a$$

have exactly two distinct real solutions?

A $a < \frac{1}{4}$

B $0 < a < \frac{1}{4}$

C $0 \leq a \leq \frac{1}{4}$

D $a \leq 0$

E $0 \leq a < \frac{1}{4}$

F $0 < a \leq \frac{1}{4}$

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 11–12

QUESTION 11

Choose one answer.

For which real values of k does

$$|x^2 - 4| = k$$

have exactly three distinct real solutions?

A $k = 0$

B $k = 2$

C $k = 4$

D $0 < k < 4$

E $k > 4$

F No real value of k

Working

QUESTION 12

Choose one answer.

The function g is defined on its natural real domain by

$$g(x) = \frac{x^2 - 4}{x^2 - x - 2}.$$

What is the range of g? Here $\mathbb{R}$ denotes the set of real numbers.

A $\mathbb{R} \setminus \{1\}$

B $\mathbb{R} \setminus \{1, \frac{4}{3}\}$

C $\mathbb{R} \setminus \{-1, 2\}$

D $\mathbb{R} \setminus \{1, 2\}$

E $\mathbb{R} \setminus \{-1, \frac{4}{3}\}$

F $\mathbb{R}$

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Questions 13–14

QUESTION 13

Choose one answer.

The function f is defined for all real x by

$$f(x) = (x-2)^2 - 3.$$

The graphs of $y = f(x)$ and $y = f(2x+1)$ intersect at two points.

What is the sum of the y-coordinates of these points?

A -4

B -2

C 0

D 2

E 4

F 8

Working

QUESTION 14

Choose one answer.

A student attempts to solve

$$(x-1)\sqrt{x+2} = (x-1)x.$$

Step I: divide by $x-1$, giving $\sqrt{x+2} = x$.
Step II: square both sides, giving $x+2 = x^2$.

The student concludes that $x = -1$ or $x = 2$. Which statement is correct?

A Both steps preserve exactly the original solution set.

B Step I loses $x = 1$; step II introduces no extra candidate.

C Step I loses no solution; step II introduces $x = -1$.

D Step I loses $x = 1$; step II introduces $x = -1$.

E Step I loses $x = 1$; step II introduces $x = 2$.

F Step I loses $x = -1$; step II introduces $x = 1$.

Working

PRACTICE · MEDIUM TO MEDIUM-HARD

Question 15 & review

QUESTION 15 Choose one answer.

Let a be a real constant. Consider the statements

> S: $x^2 - 2ax + 1 > 0$ for every real $x \geq 0$.
>
> T: $-1 < a < 1$.

Which statement correctly describes T as a condition for S?

A T is both necessary and sufficient.

B T is sufficient but not necessary.

C T is necessary but not sufficient.

D T is neither necessary nor sufficient.

Working

Second-pass review

Which two questions would you revisit first, and why?

Write one restriction or boundary case you initially missed.

TEACHING EDITION · ANSWERS START HERE

Answer key

Use this page only after an independent attempt. Each question has one correct option. The worked answers explain both the method and the restriction that makes it valid.

Q	ANSWER	RESULT	MAIN FOCUS
01	C	18	Fractional indices and reciprocal structure
02	D	14	Exact surd inequalities and integer counting
03	E	$5 < k < 9$	Location of two distinct quadratic roots
04	A	$1 \leq k < 4$	Global positivity and a disappearing quadratic term
05	B	-5	Line–parabola intersections and root information
06	F	$\{-1\} \cup (1, 2) \cup (2, \infty)$	Rational inequalities: holes and isolated solutions
07	A	$2x + 3$	Remainder theorem with non-monic divisors
08	D	$7x + 3$	Division by a quadratic and remainder reduction
09	C	3	Composite functions and distinct preimages
10	E	$0 \leq a < \frac{1}{4}$	Radical substitution and admissible roots
11	C	$k = 4$	Modulus equations and boundary root counts
12	B	$\mathbb{R} \setminus \{1, \frac{4}{3}\}$	Range of a rational function with a cancelled hole
13	E	4	Transformed quadratics and equal outputs
14	D	Lost: 1; introduced: -1	Proof error: a lost root and an extraneous root
15	B	Sufficient, not necessary	Sufficient versus necessary on a restricted domain

A useful review rule. A correct option reached by an invalid step still needs review. In particular, check the original domain, the sign required before squaring, and whether the question asks for distinct roots.

The questions are deliberately paired by topic with questions 1–15 in the original challenge pack. No official scaled score should be inferred from this topic set.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 1–2

01 · Answer C

Fractional indices and reciprocal structure

Key observation. The fractional powers become a reciprocal pair after one substitution.

Put $t = x^{1/3} > 0$. Then $t^2 + t^{-2} = 7$, so

$$\left(t + \frac{1}{t}\right)^2 = 9 \quad \Longrightarrow \quad t + \frac{1}{t} = 3.$$

The negative square root is impossible because $t > 0$. The equation $t^2 - 3t + 1 = 0$ has two distinct positive roots, whose product is 1. Call them u and u^{-1}.

The two possible original values are therefore u^3 and u^{-3}. Their sum is

$$u^3 + u^{-3} = \left(u + u^{-1}\right)^3 - 3\left(u + u^{-1}\right) = 27 - 9 = 18.$$

Both values are admissible and distinct, so the answer is **C**.

> **Trap to avoid.** Do not stop at the two values of t: the question asks about $x = t^3$.

---

02 · Answer D

Exact surd inequalities and integer counting

Key observation. Rearrange before squaring, so both sides are known to be positive.

For $n \geq 0$, the given inequality is equivalent to

$$\sqrt{n+5} > \sqrt{n+1} + \frac{1}{2}.$$

Both sides are positive. Squaring is therefore reversible:

$$n + 5 > n + 1 + \sqrt{n+1} + \frac{1}{4} \quad \Longleftrightarrow \quad \sqrt{n+1} < \frac{15}{4}.$$

Squaring these non-negative quantities gives

$$n + 1 < \frac{225}{16} \quad \Longleftrightarrow \quad n < \frac{209}{16} = 13 + \frac{1}{16}.$$

Thus n can be any of $0, 1, \ldots, 13$: **14 integers**, giving **D**.

Rationalising also gives $4/(\sqrt{n+5} + \sqrt{n+1})$, which explains why the expression decreases as n increases.

> **Trap to avoid.** Non-negative integers include 0. Counting only 1 through 13 loses one valid value.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 3–4

03 · Answer E

Location of two distinct quadratic roots

Key observation. Use the smaller root to enforce the location condition.

Completing the square gives

$$(x-3)^2 = 9-k.$$

There are two distinct real roots exactly when $k<9$. They are $3 \pm \sqrt{9-k}$.

Both exceed 1 precisely when the smaller one does:

$$3-\sqrt{9-k} > 1 \quad \Longleftrightarrow \quad \sqrt{9-k} < 2.$$

For $k<9$, this gives $9-k<4$, or $k>5$. Combining the conditions,

$$\boxed{5<k<9}.$$

At $k=5$, one root is 1, which is not allowed. At $k=9$, the two roots coincide. Hence **E**.

> **Trap to avoid.** A positive discriminant establishes two real roots, but says nothing by itself about whether both exceed 1.

---

04 · Answer A

Global positivity and a disappearing quadratic term

Key observation. Treat the zero leading coefficient separately, then use the minimum.

Rewrite the expression as

$$(k-1)(x+1)^2 + 4 - k.$$

If $k<1$: the squared term has a negative coefficient, so the expression becomes negative for sufficiently large $|x|$.

If $k=1$: the expression is the constant 3. This case satisfies the strict inequality for every x.

If $k>1$: the minimum occurs at $x=-1$ and is $4-k$. Strict positivity requires $4-k>0$, so $k<4$.

Combining the successful cases gives $1 \le k < 4$, which is **A**. At $k=4$, the expression is zero at $x=-1$, so the upper endpoint is excluded.

> **Trap to avoid.** The usual quadratic test assumes a non-zero leading coefficient. It must not discard the valid constant case $k=1$.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 5–6

05 · Answer B

Line–parabola intersections and root information

Key observation. The midpoint and separation determine the two intersection inputs immediately.

The two x-coordinates are

$$3 - \frac{4}{2} = 1 \qquad \text{and} \qquad 3 + \frac{4}{2} = 5.$$

Equating the line and parabola gives

$$x^2 - (m+2)x - c = 0.$$

This monic quadratic has roots 1 and 5, so it equals

$$(x-1)(x-5) = x^2 - 6x + 5.$$

Comparing constant terms gives $-c = 5$, hence $c = -5$: **B**. As a check, comparing linear coefficients gives $m = 4$; the line passes through $(1, -1)$ and $(5, 15)$.

> **Trap to avoid.** The constant term of the intersection equation is $-c$, not c. The difference of 4 is between the inputs, not the outputs.

---

06 · Answer F

Rational inequalities: holes and isolated solutions

Key observation. Keep both original exclusions and check where the squared numerator is zero.

The original expression is undefined at $x = 1$ and $x = 2$. For all other inputs it simplifies to

$$\frac{(x+1)^2}{x-1}.$$

For $x > 1$, the numerator and denominator are positive, so the inequality holds, except at the original excluded input $x = 2$.

For $x < 1$, the denominator is negative. The quotient is negative unless the numerator is zero. This happens at $x = -1$, which is permitted and gives equality.

Therefore the complete set is

$$\boxed{\{-1\} \cup (1, 2) \cup (2, \infty)},$$

giving **F**.

> **Trap to avoid.** Cancelling $x - 2$ does not restore the input 2. The equality sign also makes -1 a valid isolated solution.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 7–8

07 · Answer A

Remainder theorem with non-monic divisors

Key observation. The new divisor contains both of the original linear factors.

Factor the quadratic divisor:

$$2x^2 + x - 1 = (2x - 1)(x + 1).$$

Its remainder has degree less than 2, so write it as $R(x) = ax + b$. At either zero of the divisor, P and R have the same value.

The given remainders mean $P(\frac{1}{2}) = 4$ and $P(-1) = 1$. Hence

$$\frac{a}{2} + b = 4, \qquad -a + b = 1.$$

Subtracting gives $3a/2 = 3$, so $a = 2$ and $b = 3$. The required remainder is $2x + 3$: **A**.

> **Trap to avoid.** Division by $2x - 1$ gives information at $x = \frac{1}{2}$, not at $x = 1$. A quadratic divisor can leave a linear remainder.

---

08 · Answer D

Division by a quadratic and remainder reduction

Key observation. The divisor factorises, so two evaluations determine the linear remainder.

Let $P(x) = 4x^4 + 2x^3 + 3x + 1$. Since

$$2x^2 - x - 1 = (2x + 1)(x - 1),$$

a remainder $ax + b$ must satisfy

$$a + b = P(1) = 10, \qquad -\frac{a}{2} + b = P\left(-\frac{1}{2}\right) = -\frac{1}{2}.$$

Subtracting gives $3a/2 = 21/2$, so $a = 7$ and $b = 3$. Thus **D** is correct.

Division check. Direct polynomial division gives the identity

$$4x^4 + 2x^3 + 3x + 1 = (2x^2 - x - 1)(2x^2 + 2x + 2) + 7x + 3.$$

The remainder has degree 1, which is lower than the divisor's degree 2.

> **Trap to avoid.** Include the missing $0x^2$ term when using long division. Stopping with a quadratic remainder is not valid.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 9–10

09 · Answer C

Composite functions and distinct preimages

Key observation. Solve for the intermediate output before finding the original inputs.

Put $u = f(x)$. The outer equation becomes

$$u^2 - 2u = 3 \quad \Longleftrightarrow \quad (u-3)(u+1) = 0.$$

Thus either $f(x) = 3$ or $f(x) = -1$.

For $f(x) = 3$, we have $x^2 - 2x - 3 = 0$, giving $x = -1$ or $x = 3$.

For $f(x) = -1$, we have $x^2 - 2x + 1 = 0$, giving only $x = 1$.

The complete set is $\{-1, 1, 3\}$, containing **three distinct values**. Hence **C**.

Equivalently, $f(f(x)) - 3 = (x-1)^2(x+1)(x-3)$. The double factor at 1 does not give a second distinct input.

> **Trap to avoid.** A quadratic outer equation does not guarantee two inputs for each output. A turning-point output has only one preimage.

---

10 · Answer E

Radical substitution and admissible roots

Key observation. The substitution must retain the non-negative sign of the principal square root.

Put $t = \sqrt{x}$, so $t \geq 0$ and $x = t^2$. The equation becomes

$$t^2 - t + a = 0.$$

Each non-negative value of t corresponds to exactly one original x. We therefore need two distinct non-negative roots in t.

Distinct real roots require $1 - 4a > 0$, or $a < \frac{1}{4}$. Their sum is 1 and their product is a. Given two real roots with positive sum, both are non-negative exactly when their product is non-negative. Hence $a \geq 0$.

Combining gives $0 \leq a < \frac{1}{4}$: **E**. At $a = 0$, the original solutions are 0 and 1. At $a = \frac{1}{4}$, there is only one distinct solution, $x = \frac{1}{4}$.

> **Trap to avoid.** The discriminant alone can count a negative t, which is impossible for $\sqrt{x}$. The endpoint $a = 0$ is allowed.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 11–12

11 · Answer C

Modulus equations and boundary root counts

Key observation. Three roots occur when one pair merges at zero while the other pair remains.

For $k < 0$, there are no solutions. For $k \geq 0$, the equation is equivalent to

$$x^2 = 4 + k \qquad \text{or} \qquad x^2 = 4 - k.$$

PARAMETER	NUMBER OF DISTINCT REAL SOLUTIONS
$k = 0$	2: both branches give the same two roots.
$0 < k < 4$	4: two different positive squares, each with two roots.
$k = 4$	3: $x = \pm 2\sqrt{2}$ and $x = 0$.
$k > 4$	2: the second branch has no real roots.

Exactly three distinct roots occur only at $k = 4$, giving **C**.

> **Trap to avoid.** The equation $x^2 = 0$ has one distinct root, not two. At $k = 0$, do not count the same two roots twice.

---

12 · Answer B

Range of a rational function with a cancelled hole

Key observation. There is an unattainable output from the rational form and another from the cancelled hole.

Factor first, retaining the original domain:

$$g(x) = \frac{(x-2)(x+2)}{(x-2)(x+1)} = \frac{x+2}{x+1}, \qquad x \neq -1, 2.$$

Set $y = (x+2)/(x+1)$. Rearranging gives

$$x = \frac{2-y}{y-1}.$$

The value $y = 1$ is impossible. For every other y, the formula gives a unique input and never gives $x = -1$.

The other excluded input, $x = 2$, corresponds to $y = 4/3$. Because the input for each output is unique, no alternative input restores this value.

Every real output other than 1 and $4/3$ is attainable. The range is $\mathbb{R} \setminus \{1, \frac{4}{3}\}$, so **B**.

> **Trap to avoid.** Domain exclusions are inputs, not outputs. Cancelling a factor can remove an attainable-looking output from the simplified range.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answers 13–14

13 · Answer E

Transformed quadratics and equal outputs

Key observation. Equal squares give two branches, not just equal expressions inside the squares.

At an intersection,

$$(x-2)^2-3=(2x-1)^2-3,$$

so $(x-2)^2=(2x-1)^2$. Thus either

$$x-2=2x-1 \quad \text{or} \quad x-2=-(2x-1).$$

The first gives $x=-1$; the second gives $x=1$. The corresponding outputs are

$$f(-1)=9-3=6, \qquad f(1)=1-3=-2.$$

The required sum is $6+(-2)=4$, giving **E**.

> **Trap to avoid.** A quadratic is not one-to-one over all real inputs. Equating only the expressions inside the squares loses one intersection.

---

14 · Answer D

Proof error: a lost root and an extraneous root

Key observation. Check the cancelled factor first, then the sign needed before squaring.

The original domain is $x \geq -2$. The value $x=1$ satisfies the original equation because both sides are zero. Dividing by $x-1$ without a separate case therefore loses this solution.

For $x \neq 1$, the reduced equation is $\sqrt{x+2}=x$. Its left side is non-negative, so any solution must have $x \geq 0$.

Squaring produces

$$x^2-x-2=(x-2)(x+1)=0.$$

The candidate $x=2$ is valid. The candidate $x=-1$ violates the sign condition and fails the unsquared equation.

Thus step I loses 1 and step II introduces -1: **D**. The actual original solution set is $\{1,2\}$.

> **Trap to avoid.** Squaring gives a valid forward implication. The error is treating it as reversible without checking the original sign condition.

SOLUTIONS · METHOD, RESTRICTIONS & CHECKS

Worked answer 15 & teaching notes

15 · Answer B

Sufficient versus necessary on a restricted domain

Key observation. Prove sufficiency, then use a negative parameter to test necessity.

If $-1 < a < 1$, then $1 - a^2 > 0$. Completing the square gives

$$x^2 - 2ax + 1 = (x - a)^2 + 1 - a^2 > 0$$

for every real x, and therefore for every $x \geq 0$. Hence T is sufficient for S.

It is not necessary. Take $a = -2$, for which T is false. On $x \geq 0$,

$$x^2 - 2ax + 1 = x^2 + 4x + 1 \geq 1 > 0.$$

So S is true even though T is false. The answer is **B**.

Exact condition. In fact S holds precisely when $a < 1$. If $a \leq 0$, each term in $x^2 - 2ax + 1$ is non-negative and the constant is positive. If $0 < a < 1$, use the completed square. If $a \geq 1$, the allowed input $x = a$ gives $1 - a^2 \leq 0$.

> **Trap to avoid.** A condition that guarantees positivity for all real inputs can be stronger than necessary on the half-line $x \geq 0$.

Moving on to the harder pack

Use the same-numbered challenge question after a student can explain the decisive observation here, not merely select the right option. Questions 1, 3, 6, 7 and 14 make a useful first-session discussion set. Questions 9–13 can accompany the later functions-and-graphs session.

Style references and provenance

Official papers were consulted for concise multiple-choice wording, parameter reasoning and error-diagnosis style: TMUA 2023 Paper 1, questions 2 and 11; TMUA 2018 Paper 2, question 9. The UAT-UK archive hosts these published materials.

All 15 questions and solutions in this companion are newly written for this pack. No official question is reproduced. Scope follows the specification extract supplied for the original algebra pack. Prepared 31 October 2026.